Performance and Limitations

Area II: Technical Subject Areas
ACS Task II.F (AI.II.F)

Cessna 172S NAV III (G1000)
[Name], CFI Applicant

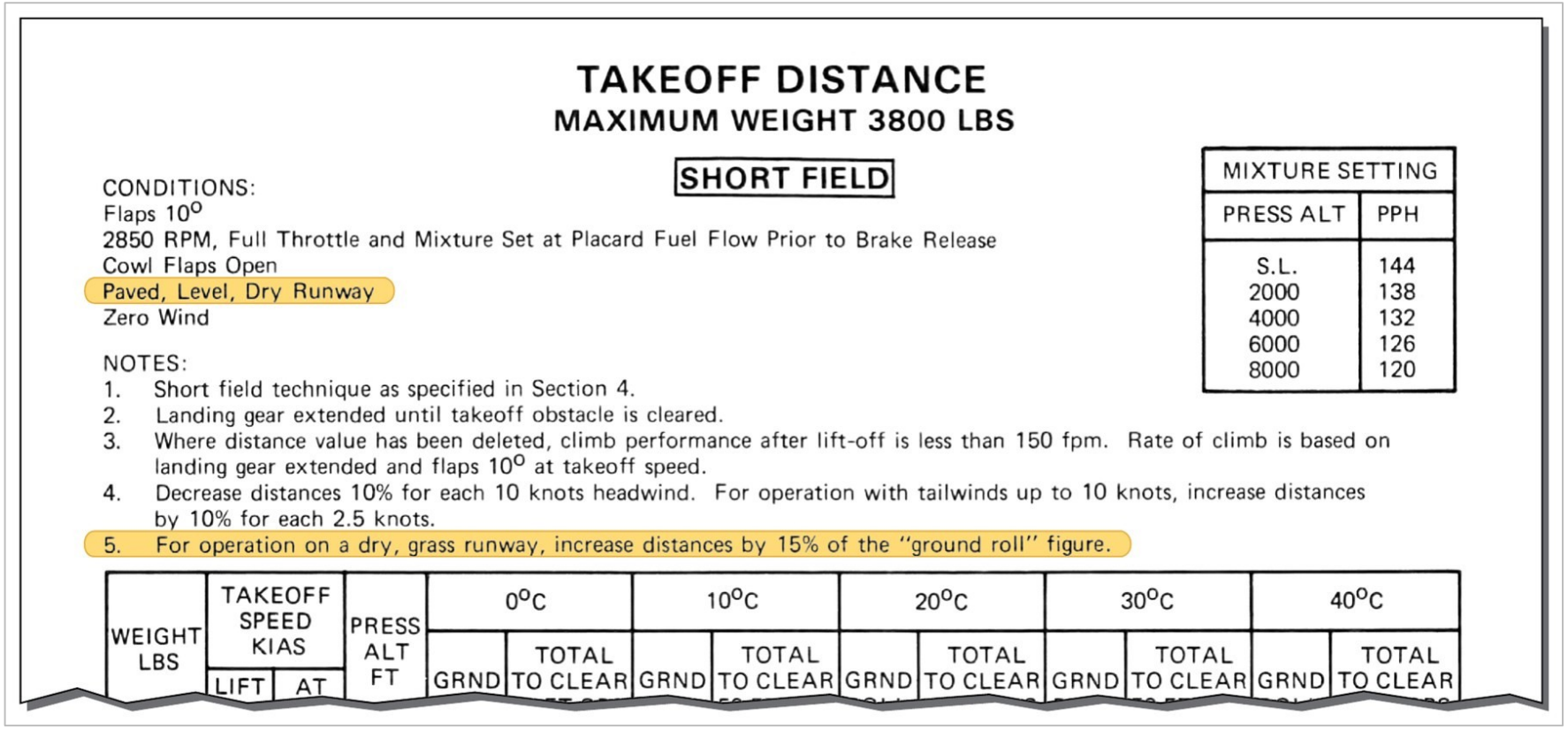




PHAK Fig 11-16

Objective

- Read POH performance charts
- Explain what changes performance
- Compute weight and CG
- Fix loading that is out of limits



TAKEOFF DISTANCE
MAXIMUM WEIGHT 3800 LBS

SHORT FIELD

CONDITIONS:
Flaps 10°
2850 RPM, Full Throttle and Mixture Set at Placard Fuel Flow Prior to Brake Release
Cowl Flaps Open
Paved, Level, Dry Runway
Zero Wind

MIXTURE SETTING	
PRESS ALT	PPH
S.L.	144
2000	138
4000	132
6000	126
8000	120

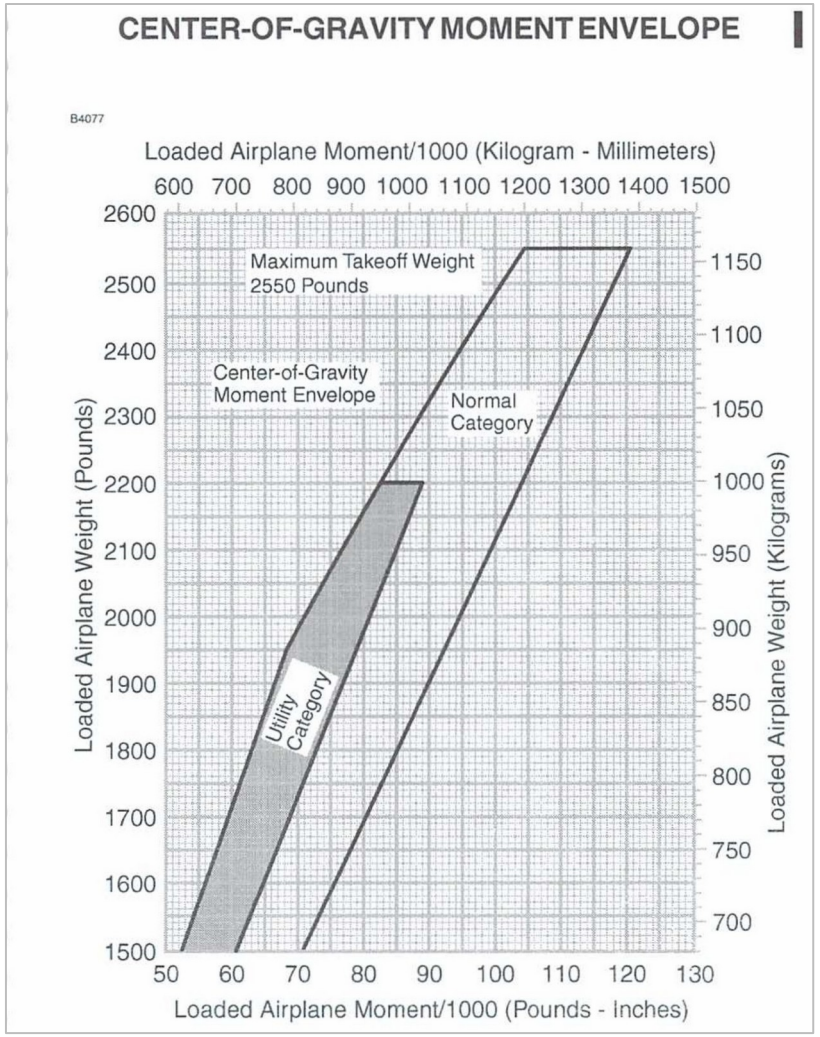
NOTES:
1. Short field technique as specified in Section 4.
2. Landing gear extended until takeoff obstacle is cleared.
3. Where distance value has been deleted, climb performance after lift-off is less than 150 fpm. Rate of climb is based on landing gear extended and flaps 10° at takeoff speed.
4. Decrease distances 10% for each 10 knots headwind. For operation with tailwinds up to 10 knots, increase distances by 10% for each 2.5 knots.
5. For operation on a dry, grass runway, increase distances by 15% of the "ground roll" figure.

WEIGHT LBS	TAKEOFF SPEED KIAS		PRESS ALT FT	0°C		10°C		20°C		30°C		40°C	
	LIFT	AT		GRND	TOTAL TO CLEAR	GRND	TOTAL TO CLEAR	GRND	TOTAL TO CLEAR	GRND	TOTAL TO CLEAR	GRND	TOTAL TO CLEAR

ACS II.F, PHAK Fig 11-15

Lesson Overview

- Performance charts and factors
- Weight and balance terms
- C172S worked examples
- Risk management



POH Fig 6-7

Where the Numbers Live

- POH Section 2: limitations
- POH Section 5: performance
- POH Section 6: weight and balance
- TCDS: certified limits

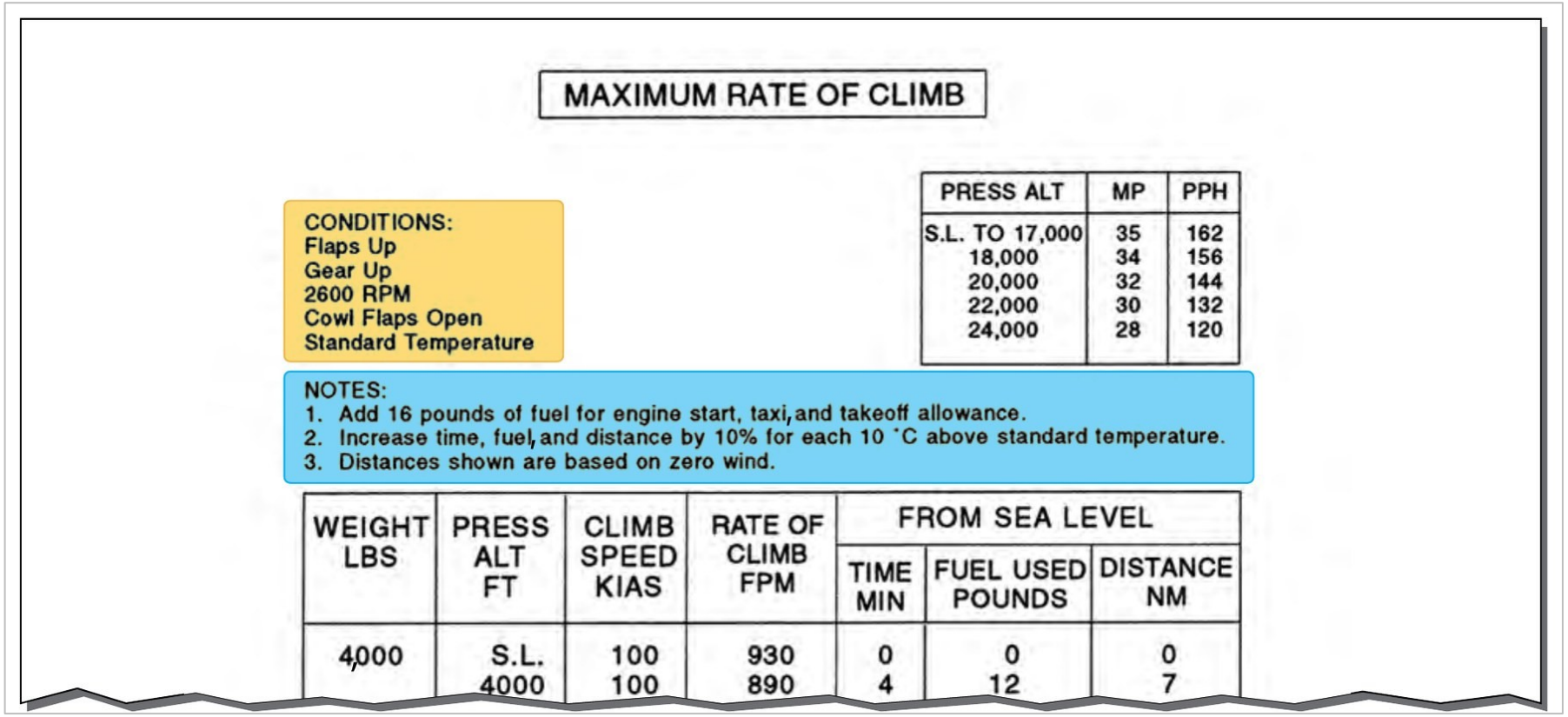
PERFORMANCE

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How the Charts Are Made

- Flight tests in a new airplane
- Average piloting skill
- Airplane in good working order
- Always read the conditions



MAXIMUM RATE OF CLIMB

CONDITIONS:
Flaps Up
Gear Up
2600 RPM
Cowl Flaps Open
Standard Temperature

PRESS ALT	MP	PPH
S.L. TO 17,000	35	162
18,000	34	156
20,000	32	144
22,000	30	132
24,000	28	120

NOTES:
1. Add 16 pounds of fuel for engine start, taxi, and takeoff allowance.
2. Increase time, fuel, and distance by 10% for each 10 °C above standard temperature.
3. Distances shown are based on zero wind.

WEIGHT LBS	PRESS ALT FT	CLIMB SPEED KIAS	RATE OF CLIMB FPM	FROM SEA LEVEL		
				TIME MIN	FUEL USED POUNDS	DISTANCE NM
4,000	S.L.	100	930	0	0	0
	4000	100	890	4	12	7

PHAK 11-19, POH 5-3, PHAK Fig 11-20

Reading Tables and Interpolating

- Find your row and column
- Interpolate between values
- Or use next higher value
- Higher value: more margin

Conditions				TAKEOFF DISTANCE MAXIMUM WEIGHT 2,400 LB									
Flaps 10° Full throttle prior to brake release Paved level runway Zero wind													
Weight (lb)	Takeoff speed KIAS		Press ALT (ft)	0 °C		10 °C		20 °C		30 °C		40 °C	
	Lift off	AT 50 ft		Grnd roll (ft)	Total feet to clear 50 ft OBS	Grnd roll (ft)	Total feet to clear 50 ft OBS	Grnd roll (ft)	Total feet to clear 50 ft OBS	Grnd roll (ft)	Total feet to clear 50 ft OBS	Grnd roll (ft)	Total feet to clear 50 ft OBS
2,400	51	56	S.L.	795	1,460	860	1,570	925	1,685	995	1,810	1,065	1,945
			1,000	875	1,605	940	1,725	1,015	1,860	1,090	2,000	1,170	2,155
			2,000	960	1,770	1,035	1,910	1,115	2,060	1,200	2,220	1,290	2,395
			3,000	1,055	1,960	1,140	2,120	1,230	2,295	1,325	2,480	1,425	2,685
			4,000	1,165	2,185	1,260	2,365	1,355	2,570	1,465	2,790	1,575	3,030
			5,000	1,285	2,445	1,390	2,660	1,500	2,895	1,620	3,160	1,745	3,455
			6,000	1,425	2,755	1,540	3,015	1,665	3,300	1,800	3,620	1,940	3,990
			7,000	1,580	3,140	1,710	3,450	1,850	3,805	2,000	4,220	- - -	- - -
			8,000	1,755	3,615	1,905	4,015	2,060	4,480	- - -	- - -	- - -	- - -

To find the takeoff distance for a pressure altitude of 2,500 feet at 20 °C, average the ground roll for 2,000 feet and 3,000 feet.

$$\frac{1{,}115 + 1{,}230}{2} = 1{,}173 \text{ feet}$$

PHAK 11-20, POH 5-4, PHAK Fig 11-21

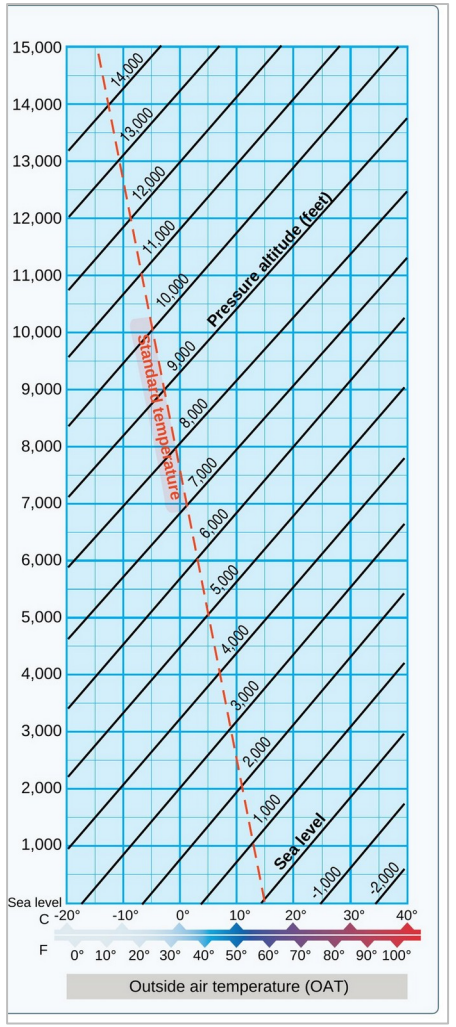
Definition: Density Altitude

Density altitude

“the altitude in the standard atmosphere corresponding to a particular value of air density.”

Also stated as

“Density altitude is pressure altitude corrected for nonstandard temperature.”



Density Altitude in Plain Words

- Hot, high, humid: thin air
- Less power, thrust, and lift
- Longer runs, weaker climbs
- Chart: hotter column, bigger number

CONDITIONS:

Flaps 10°
Full Throttle Prior to Brake Release
Paved, level, dry runway

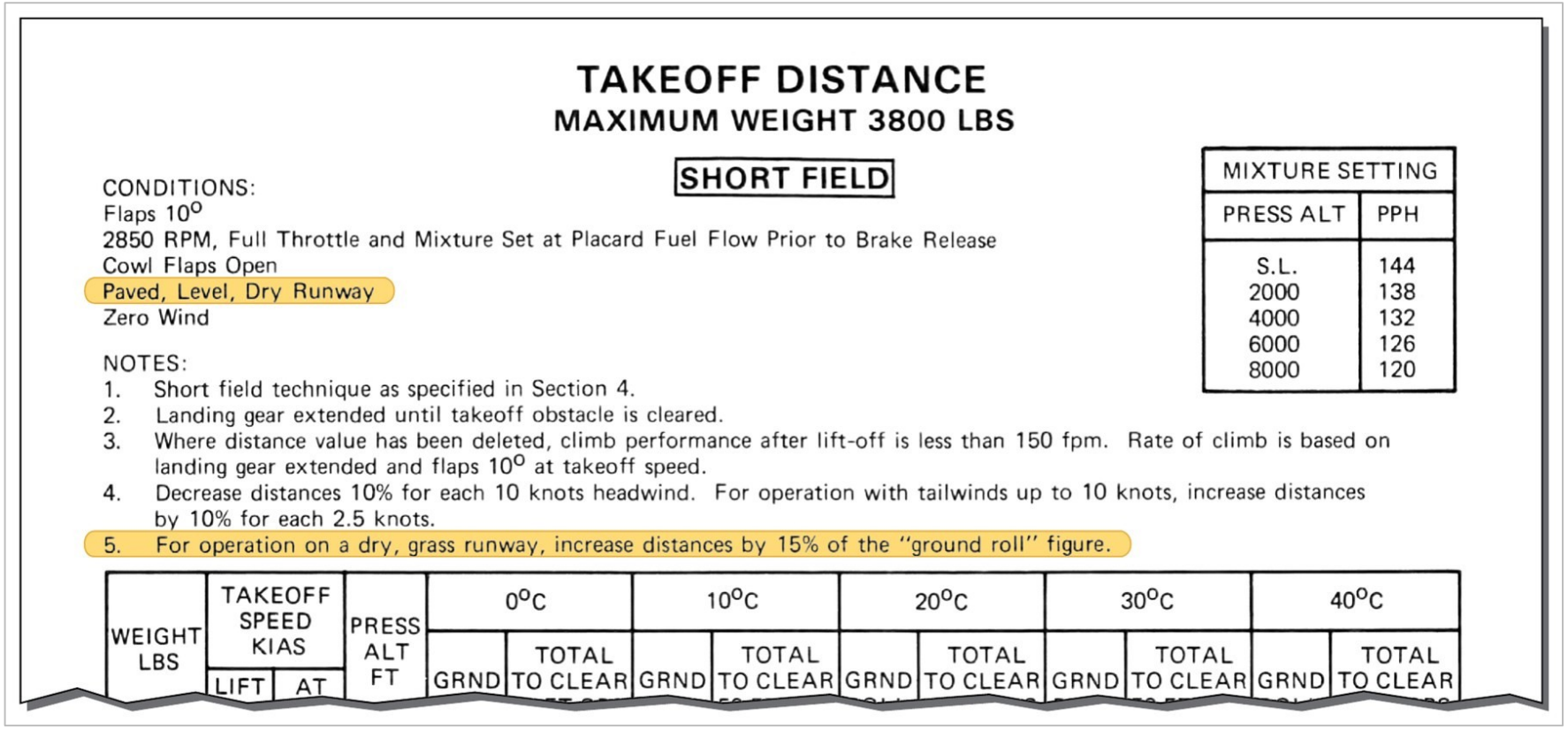
Zero Wind
Lift Off: 51 KIAS
Speed at 50 Ft: 56 KIAS

Press Alt In Feet	0°C		10°C		20°C		30°C		40°C	
	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst
S. L.	860	1465	925	1575	995	1690	1070	1810	1150	1945
1000	940	1600	1010	1720	1090	1850	1170	1990	1260	2135
2000	1025	1755	1110	1890	1195	2035	1285	2190	1380	2355
3000	1125	1925	1215	2080	1310	2240	1410	2420	1515	2605
4000	1235	2120	1335	2295	1440	2480	1550	2685	1660	2880
5000	1355	2345	1465	2545	1585	2755	1705	2975	1825	3205
6000	1495	2605	1615	2830	1745	3075	1875	3320	2010	3585
7000	1645	2910	1785	3170	1920	3440	2065	3730	2215	4045
8000	1820	3265	1970	3575	2120	3880	2280	4225	2450	4615

PHAK 11-2, PHAK 11-5, POH Fig 5-5

Pilot Technique

- Charts assume average technique
- Use the chart's exact speeds
- Short field technique: Section 4
- Weak technique: add margin



TAKEOFF DISTANCE
MAXIMUM WEIGHT 3800 LBS

SHORT FIELD

CONDITIONS:
Flaps 10°
2850 RPM, Full Throttle and Mixture Set at Placard Fuel Flow Prior to Brake Release
Cowl Flaps Open
Paved, Level, Dry Runway
Zero Wind

MIXTURE SETTING	
PRESS ALT	PPH
S.L.	144
2000	138
4000	132
6000	126
8000	120

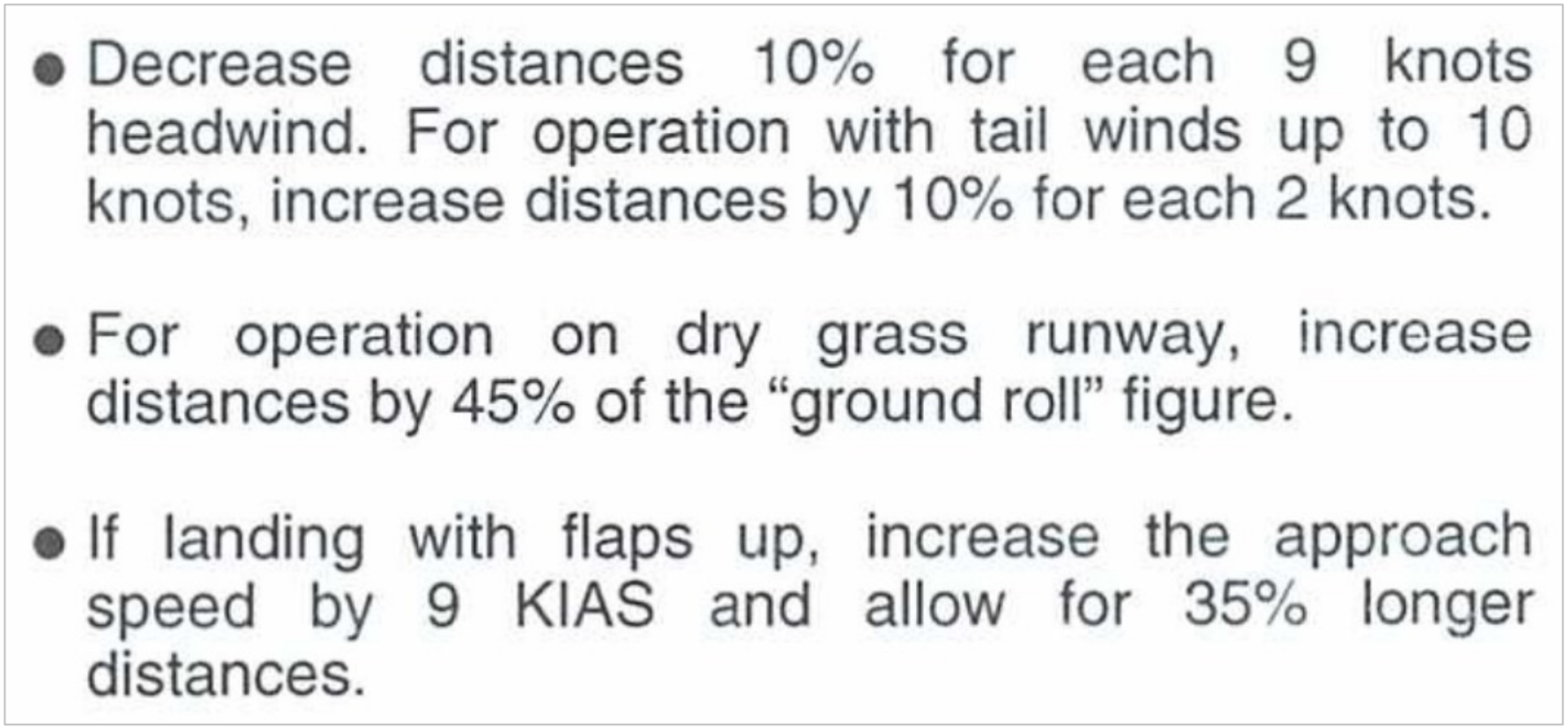
NOTES:
1. Short field technique as specified in Section 4.
2. Landing gear extended until takeoff obstacle is cleared.
3. Where distance value has been deleted, climb performance after lift-off is less than 150 fpm. Rate of climb is based on landing gear extended and flaps 10° at takeoff speed.
4. Decrease distances 10% for each 10 knots headwind. For operation with tailwinds up to 10 knots, increase distances by 10% for each 2.5 knots.
5. For operation on a dry, grass runway, increase distances by 15% of the "ground roll" figure.

WEIGHT LBS	TAKEOFF SPEED KIAS		PRESS ALT FT	0°C		10°C		20°C		30°C		40°C	
	LIFT	AT		GRND	TOTAL TO CLEAR	GRND	TOTAL TO CLEAR	GRND	TOTAL TO CLEAR	GRND	TOTAL TO CLEAR	GRND	TOTAL TO CLEAR

POH 5-3, POH Fig 5-5, PHAK 11-15, PHAK 11-19

Airplane Configuration

- Takeoff chart: flaps 10 degrees
- Climb chart: flaps up
- Flaps up landing: 35% longer
- Match the chart setup

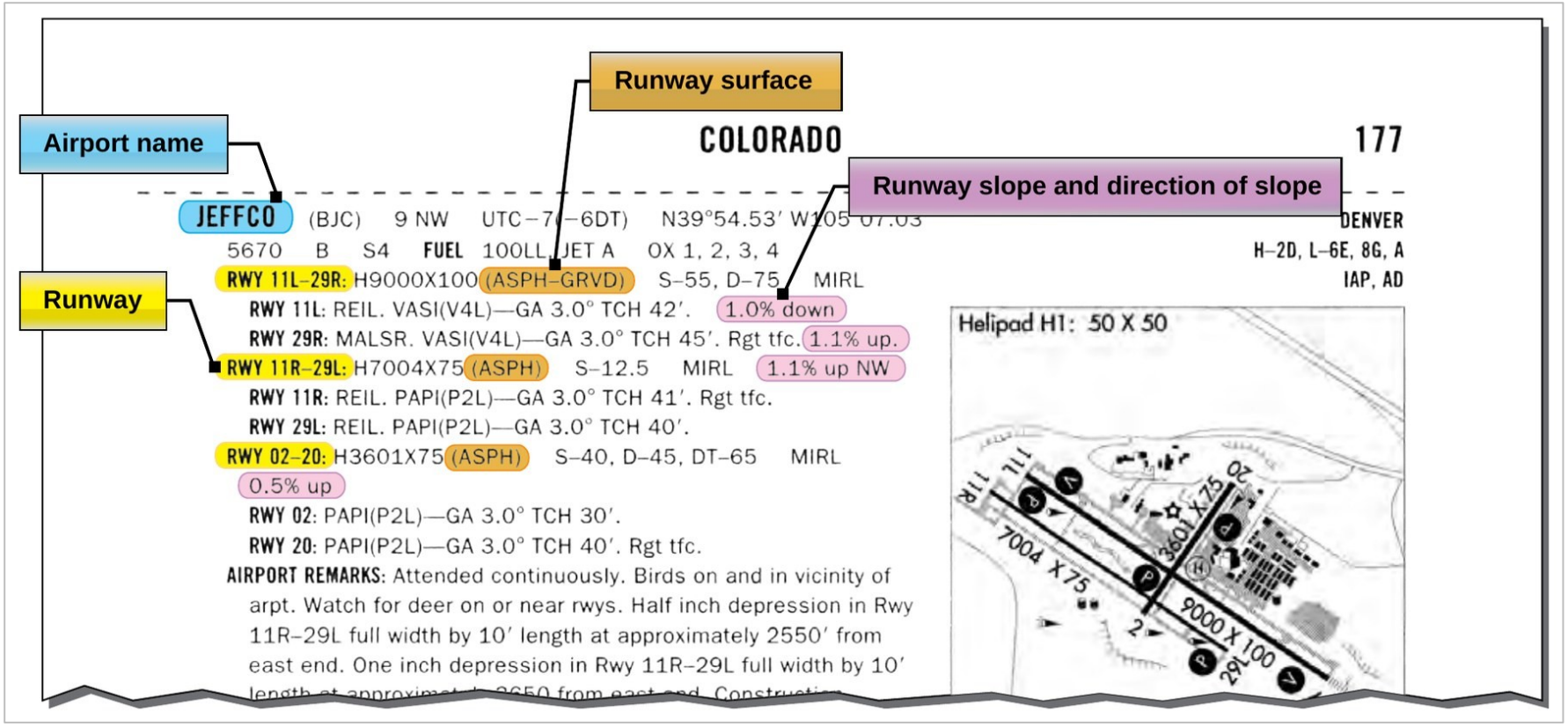


- Decrease distances 10% for each 9 knots headwind. For operation with tail winds up to 10 knots, increase distances by 10% for each 2 knots.
- For operation on dry grass runway, increase distances by 45% of the "ground roll" figure.
- If landing with flaps up, increase the approach speed by 9 KIAS and allow for 35% longer distances.

POH Fig 5-5, POH Fig 5-7, POH Fig 5-11, PHAK 11-8

Airport Environment: Surface and Slope

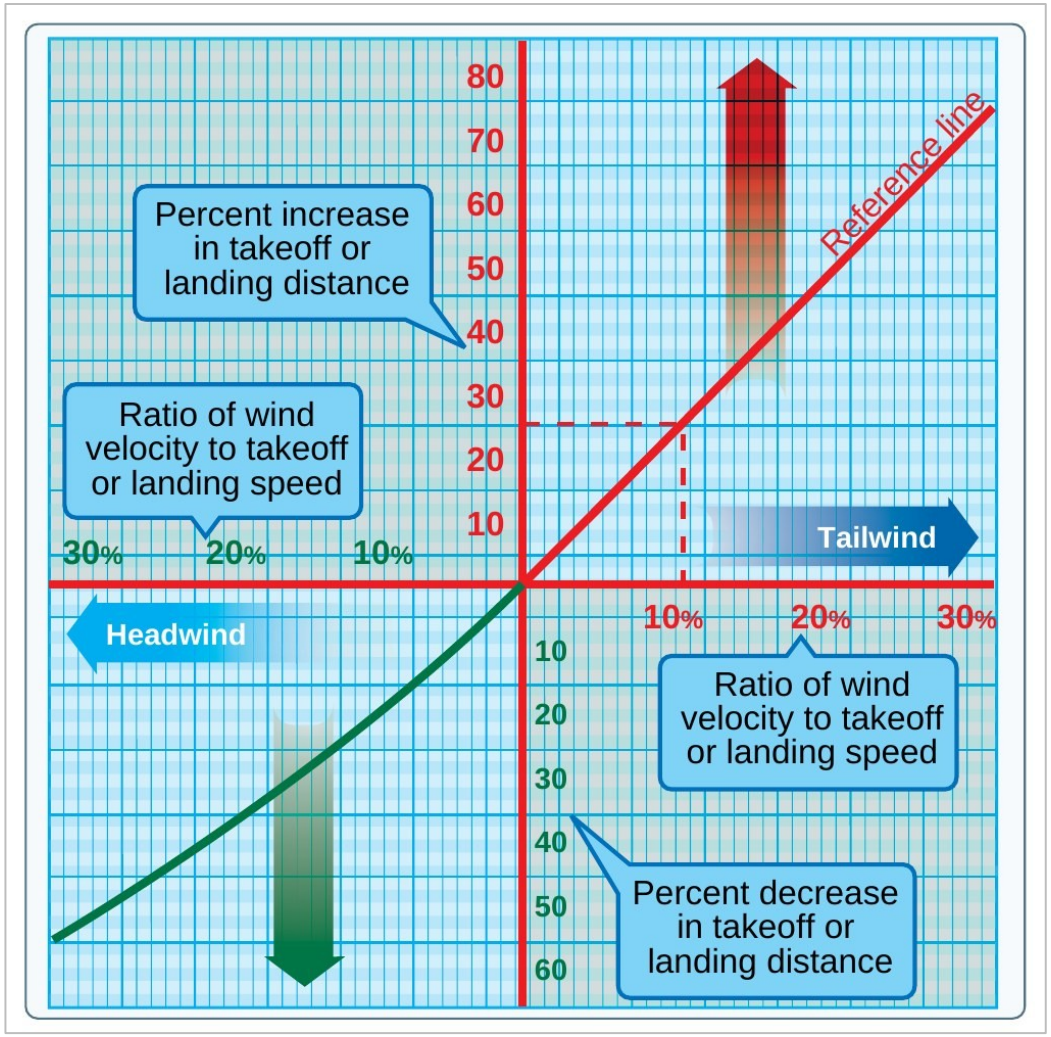
- Charts assume paved, level, dry
- Grass takeoff: +15% ground roll
- Upslope: longer takeoff roll
- Check the Chart Supplement



PHAK 11-12, PHAK 11-13, POH Fig 5-5, POH Fig 5-11, PHAK Fig 11-17

Airport Environment: Wind

- Headwind shortens distances
- Tailwind lengthens them more
- C172S: minus 10% per 9 kt
- Tailwind: plus 10% per 2 kt



PHAK 11-15, POH Fig 5-5, PHAK Fig 11-19

Loading and Weight

- Higher lift-off speed
- More mass to accelerate
- More drag and friction
- Lighter airplane: shorter runs

CONDITIONS:

Flaps 10°
Full Throttle Prior to Brake Release
Paved, level, dry runway

Zero Wind
Lift Off: 44 KIAS
Speed at 50 Ft: 50 KIAS

Press Alt In Feet	0°C		10°C		20°C		30°C		40°C	
	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst
S. L.	610	1055	655	1130	705	1205	760	1290	815	1380
1000	665	1145	720	1230	770	1315	830	1410	890	1505
2000	725	1250	785	1340	845	1435	905	1540	975	1650
3000	795	1365	860	1465	925	1570	995	1685	1065	1805
4000	870	1490	940	1605	1010	1725	1090	1855	1165	1975
5000	955	1635	1030	1765	1110	1900	1195	2035	1275	2175
6000	1050	1800	1130	1940	1220	2090	1310	2240	1400	2395
7000	1150	1985	1245	2145	1340	2305	1435	2475	1540	2650
8000	1270	2195	1370	2375	1475	2555	1580	2745	1695	2950

PHAK 11-15, PHAK 10-2, POH 5-14, POH 5-16

The Type Certificate Data Sheet

- Part of the type certificate
- Lists certified limits
- C172S: TCDS 3A12
- Engine, speeds, CG, weights

XII. Model 172S, Skyhawk SP, 4 PCLM (Normal Category), 2 PCLM (Utility Category), Approved May 1, 1998

Engine	Lycoming IO-360-L2A, Rated 180 Horsepower
Fuel	100/100LL minimum grade aviation gasoline
Engine Limits	For all operations, 2,700 RPM
Propeller and Propeller Limits	1. McCauley Propeller Hub 1A170F/JHA7660

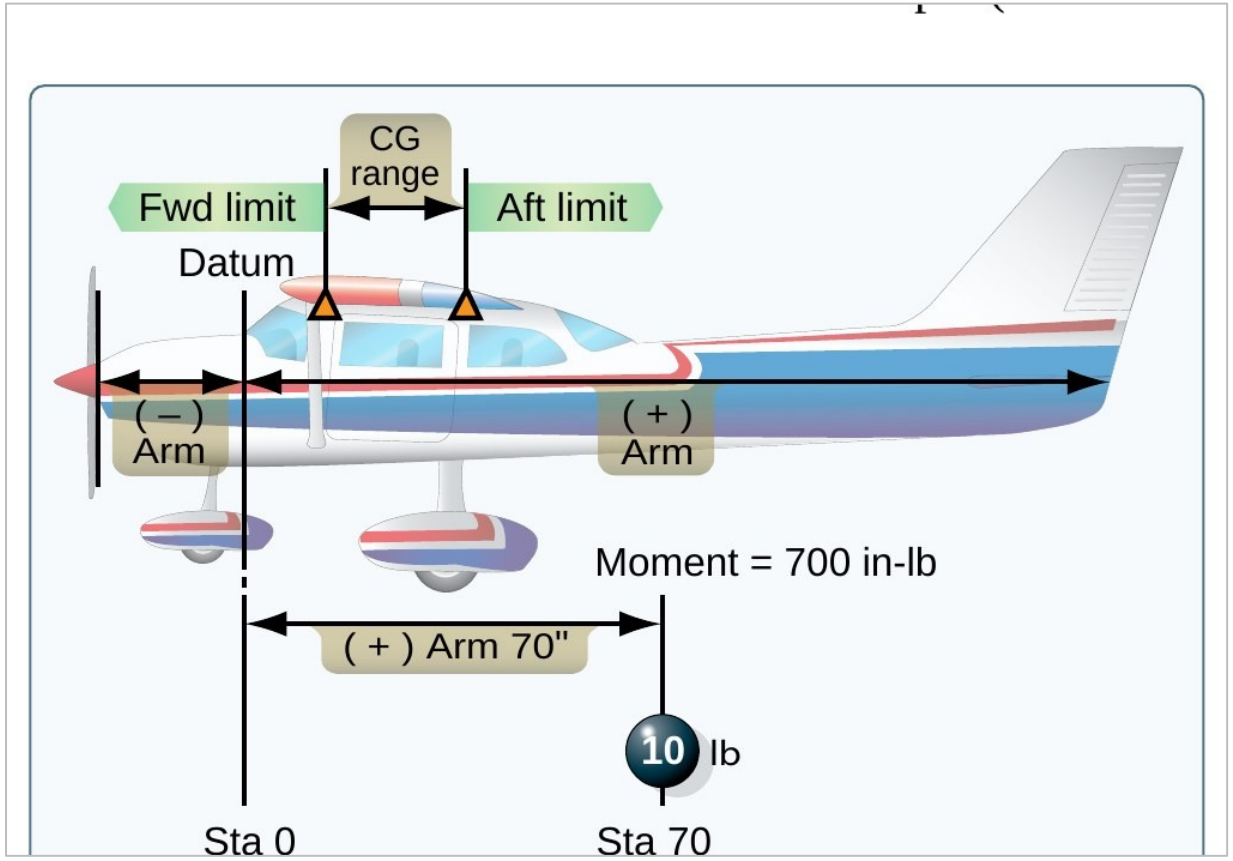
Definitions: Datum and Arm

Datum (reference datum)

“an imaginary vertical plane or line from which all measurements of arm are taken. The datum is established by the manufacturer. Once the datum has been selected, all moment arms and the location of CG range are measured from this point.”

Arm (moment arm)

“the horizontal distance in inches from the reference datum line to the CG of an item. The algebraic sign is plus (+) if measured aft of the datum and minus (−) if measured forward of the datum.”



Datum and Arm in Plain Words

- Datum: the zero point
- C172S: front face of firewall
- Arm: inches from datum
- Aft is plus, forward minus

C.G. Range	Normal Category	
	(1) Aft Limits	47.3 inches aft of datum at 2,550 pounds or less.
	(2) Forward Limits	Linear variation from 41.0 inches aft of datum at 2,550 pounds to 35.0 inches aft of datum at 1,950 pounds. 35.0 inches aft of datum at 1,950 pounds or less.
	Utility Category	
	(1) Aft Limits	40.5 inches aft of datum at 2,200 pounds or less.
	(2) Forward Limits	Linear variation from 37.5 inches aft of datum at 2,200 pounds to 35.0 inches aft of datum at 1,950 pounds. 35.0 inches aft of datum at 1,950 pounds or less.
Empty Wt. C.G. Range	None	
Reference Datum	Lower portion of front face of firewall	
MAC	58.8 inches; Leading edge of MAC 25.9 inches aft of datum	
Leveling Means	Left side of Tailcone at 108.0 inches and 142.0 inches aft of datum	

TCDS 3A12 p30, PHAK 10-6

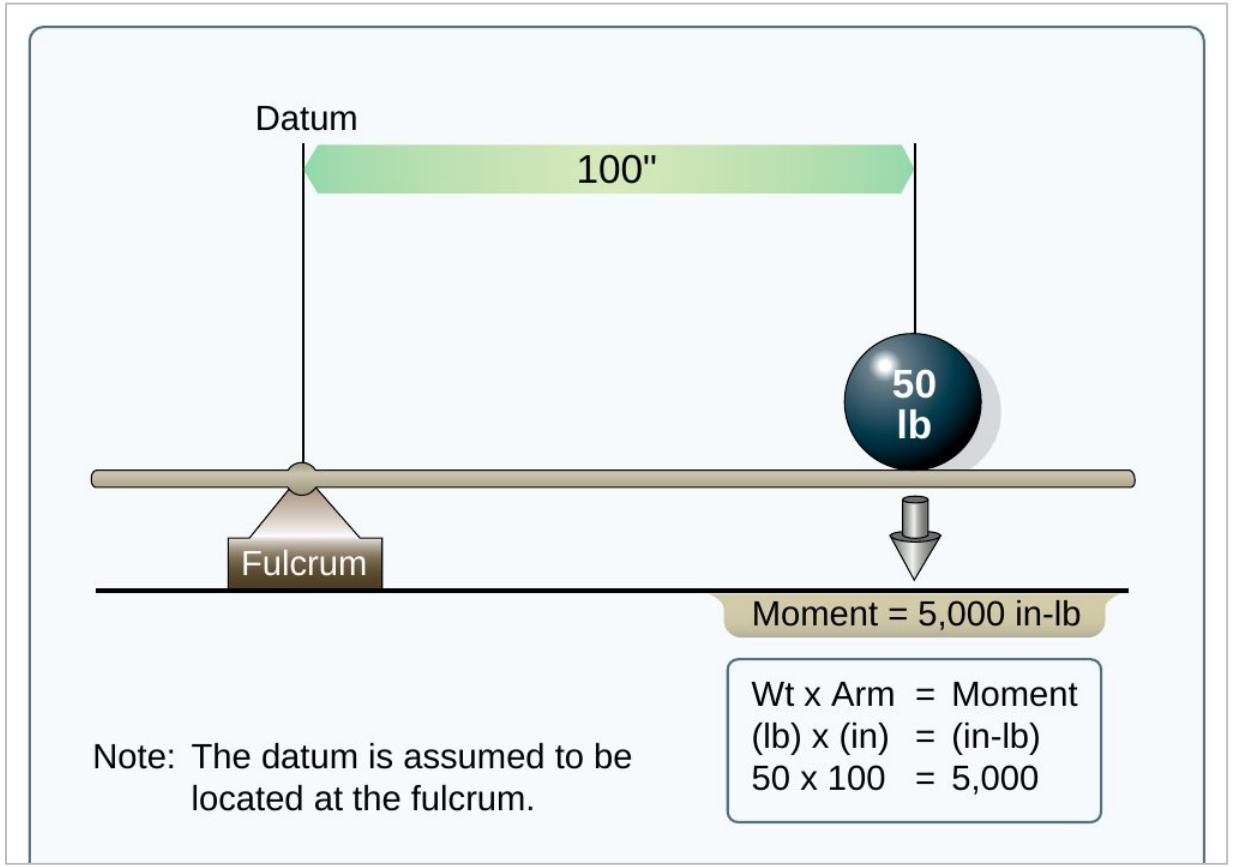
Definitions: Moment and CG

Moment

“the product of the weight of an item multiplied by its arm. Moments are expressed in pound-inches (in-lb). Total moment is the weight of the airplane multiplied by the distance between the datum and the CG.”

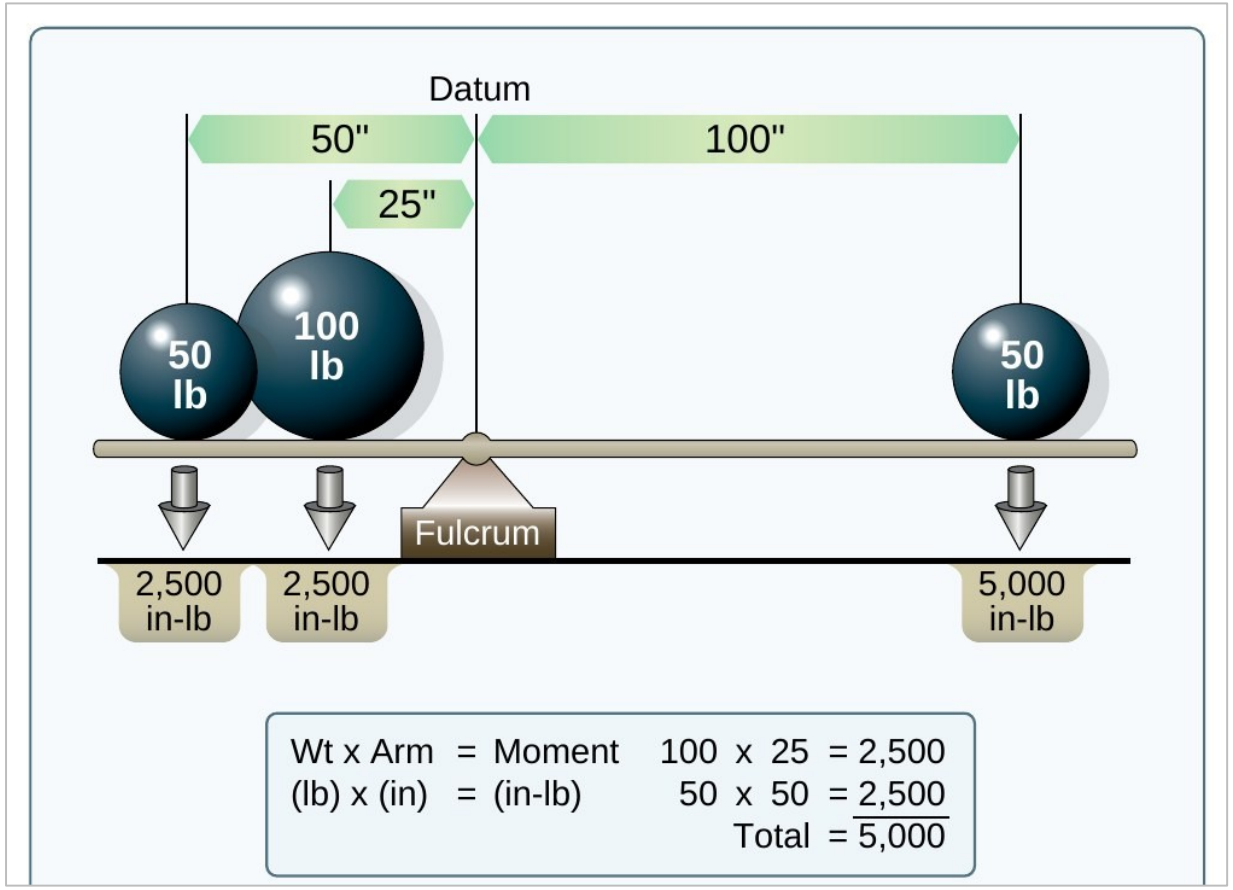
Center of gravity (CG)

“the point about which an aircraft would balance if it were possible to suspend it at that point.”



Moment and CG in Plain Words

- Moment = weight x arm
- 50 lb x 100 in = 5,000
- CG: the balance point
- CG = total moment / weight



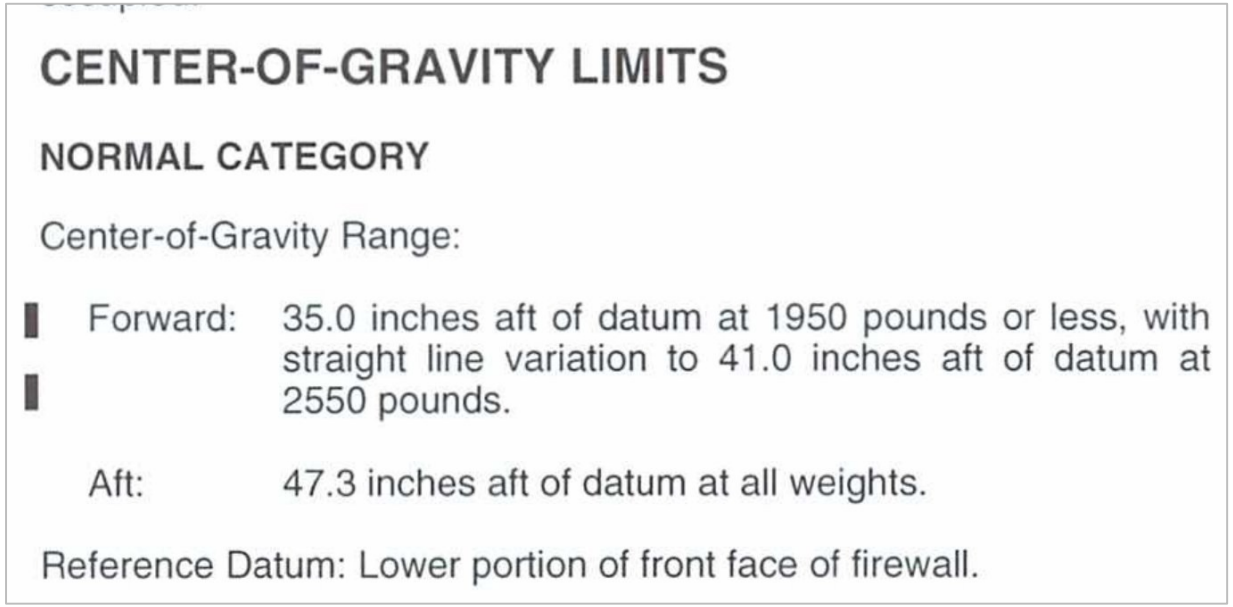
Definitions: CG Limits and Maximum Weight

CG limits

“the specified forward and aft points within which the CG must be located during flight. These limits are indicated on pertinent aircraft specifications.”

Maximum weight

“the maximum authorized weight of the aircraft and all of its equipment as specified in the TCDS for the aircraft.”



CENTER-OF-GRAVITY LIMITS

NORMAL CATEGORY

Center-of-Gravity Range:

Forward:	35.0 inches aft of datum at 1950 pounds or less, with straight line variation to 41.0 inches aft of datum at 2550 pounds.
Aft:	47.3 inches aft of datum at all weights.

Reference Datum: Lower portion of front face of firewall.

C172S Weight and CG Limits

- Max takeoff and landing: 2,550 lb
- Max ramp: 2,558 lb

- Forward: 35.0 to 41.0 in
- Aft: 47.3 in, all weights

WEIGHT LIMITS

NORMAL CATEGORY

Maximum Ramp Weight:	2558 POUNDS
Maximum Takeoff Weight:	2550 POUNDS
Maximum Landing Weight:	2550 POUNDS

Maximum Weight in Baggage Compartment:

Baggage Area A - Station 82 to 108:	120 POUNDS
Baggage Area B - Station 108 to 142:	50 POUNDS

POH 2-8, TCDS 3A12 p30

Definitions: Basic Empty Weight and Useful Load

Basic empty weight (GAMA)

“the standard empty weight plus the weight of optional and special equipment that have been installed.”

Useful load

“the weight of the pilot, copilot, passengers, baggage, usable fuel, and drainable oil. It is the basic empty weight subtracted from the maximum allowable gross weight.”

SAMPLE LOADING PROBLEM

ITEM DESCRIPTION	WEIGHT AND MOMENT TABULATION			
	SAMPLE AIRPLANE		YOUR AIRPLANE	
	Weight (Lbs.)	Moment (Lb-ins. /1000)	Weight (Lbs.)	Moment (Lb-ins. /1000)
1. Basic Empty Weight (Use the data pertaining to your airplane as it is presently equipped. Includes unusable fuel and full oil)	1642	62.6		
2. Usable Fuel (At 6 Lbs./Gal.)				
Standard Fuel 53 Gallons Maximum				
Reduced Fuel (30 Gallons)	180	8.6		
3. Pilot and Front Passenger (FS 32 to 50)	340	12.6		
4. Rear Passengers (FS 74)	340	24.8		
5.* Baggage "A" (FS 82 to 108) 120 Pounds Maximum	56	4.6		
6.* Baggage "B" (FS 108 to 142) 50 Pounds Maximum				
7. **RAMP WEIGHT AND MOMENT**	2558	113.2		
8. Fuel allowance for engine start, taxi and runup	-8.0	-0.4		
9. **TAKEOFF WEIGHT AND MOMENT** (Subtract Step 8 from Step 7)	2550	112.8		

10. Locate this point (2550 at 112.8) on the Center-of-Gravity Moment Envelope, and since this point falls within the envelope, the loading is acceptable.

* The maximum allowable combined weight capacity for baggage in areas A and B is 120 pounds.

PHAK 10-4, PHAK 10-5, POH Fig 6-3

Empty Weight and Useful Load in Plain Words

- BEW: your airplane's records
- Includes unusable fuel, full oil
- Useful load: max minus BEW
- Sample: 2,550 minus 1,642 = 908

Maximum Weights	Normal Category	
	Maximum Ramp	2,558 lb.
	Maximum Takeoff and Landing	2,550 lb.
	Utility Category	
	Maximum Ramp	2,208 lb.
	Maximum Takeoff and Landing	2,200 lb.
No. of Seats	4 (2 at 34.0 to 46.0 inches aft of datum; 2 at 73.0 inches aft of datum)	
Maximum Baggage	120 lb. at 82.0 to 108.0 inches aft of datum 50 lb. at 108.0 to 142.0 inches aft of datum (Max. combined weight capacity for baggage areas is 120 pounds)	
Fuel Capacity (Gal.)	56 gallons total; 53 gallons usable (two 28-gallon tanks in wings at 48.0 inches aft of datum). (See NOTE 1 for data on unusable fuel).	
Oil Capacity (Gal.)	8.0 quarts at 13.1 inches forward of datum 3.0 quarts usable	

POH 6-10, TCDS 3A12 p30, TCDS 3A12 p32, PHAK 10-6

Three Ways to Compute CG

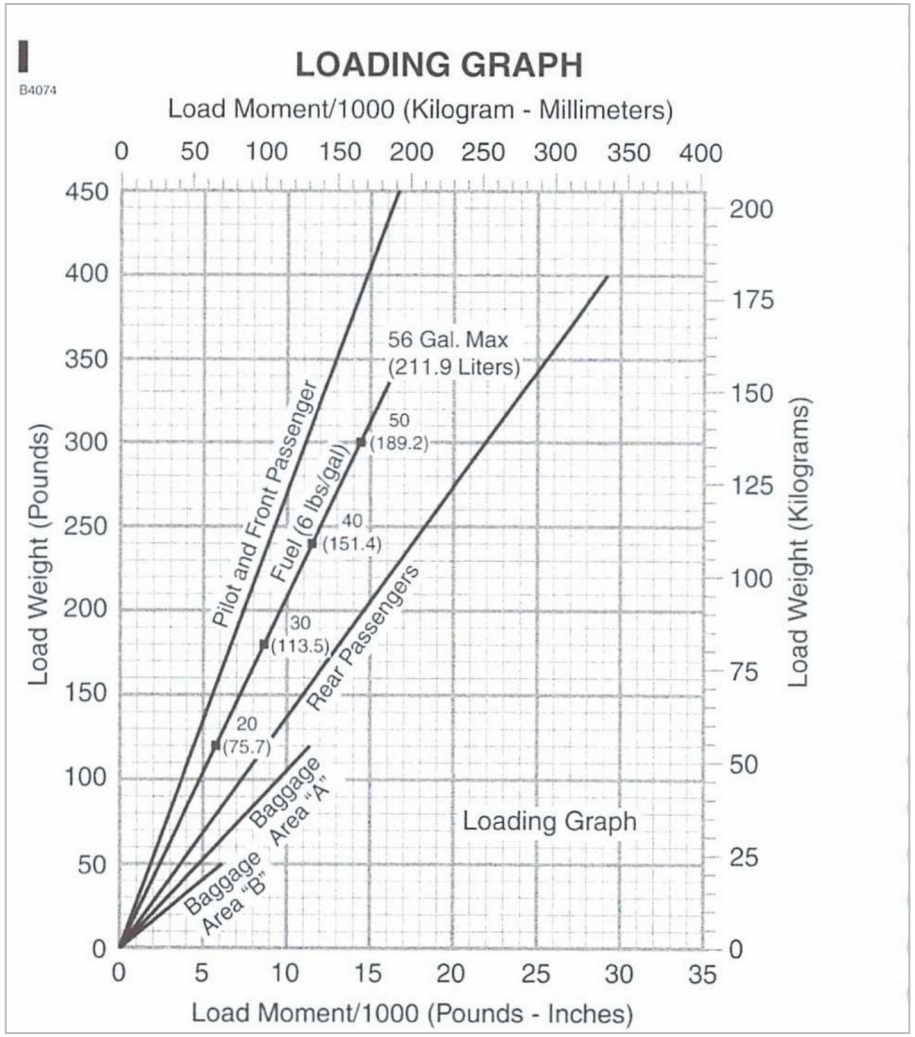
- Computation: weight x arm
- Graph: loading graph, envelope
- Table: moments already figured
- Same answer every way

Item	Weight	Arm	Moment
Aircraft Empty Weight	2,100	78.3	164,430
Front Seat Occupants	340	85.0	28,900
Rear Seat Occupants	350	121.0	42,350
Fuel	450	75.0	33,750
Baggage Area 1	80	150.0	12,000
Total	3,320		281,430
			281,430 ÷ 3,320 = 84.8

PHAK 10-7, PHAK 10-9, PHAK Fig 10-5

C172S Graph Method

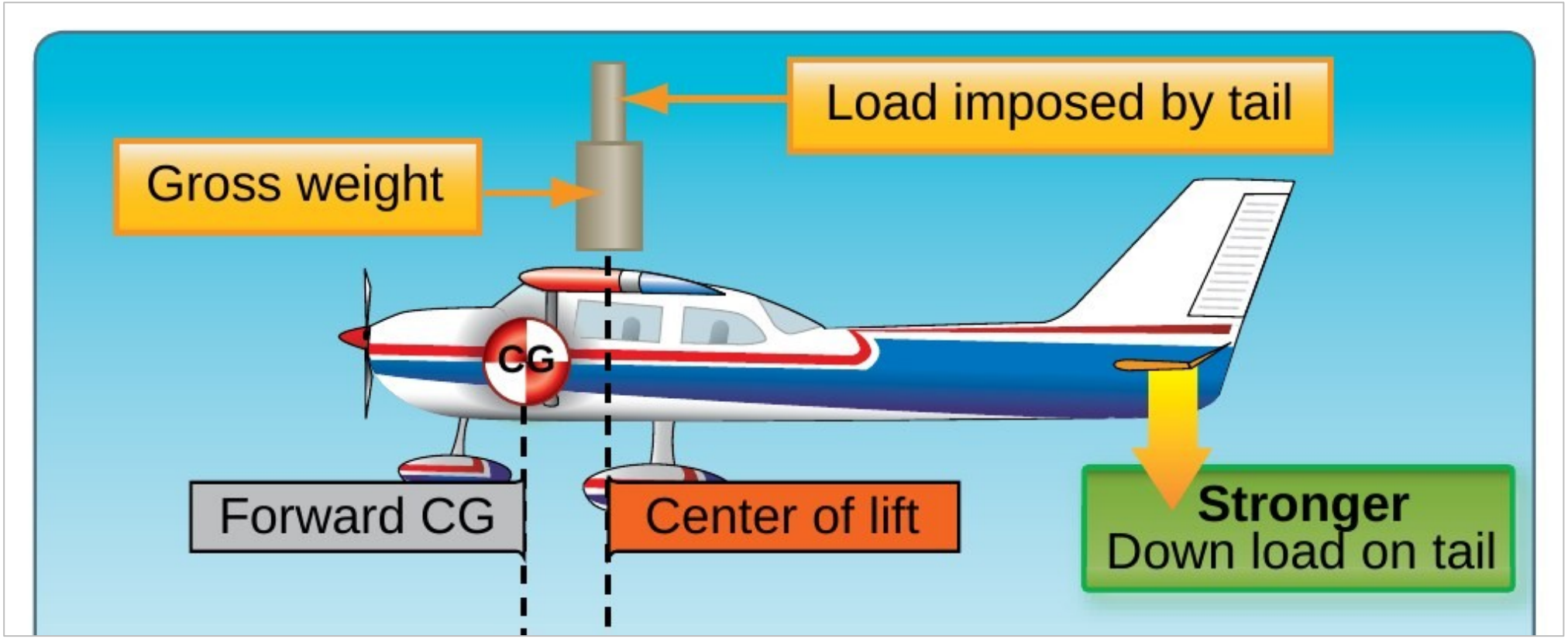
- Find load weight on left
- Go across to the item line
- Drop down: moment / 1000
- Add up, then plot envelope



PHAK 10-7, POH Fig 6-4

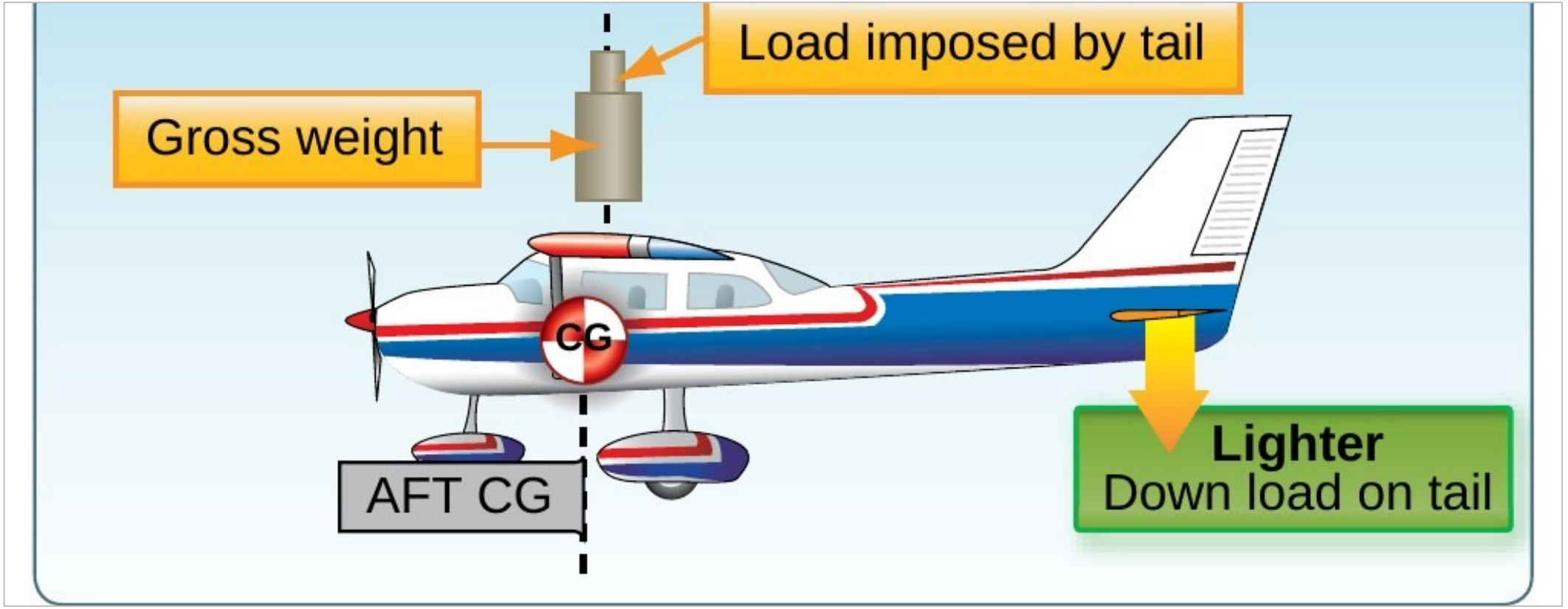
Aerodynamics: Forward CG

- Tail pushes down harder
- Higher stall speed
- Heavier elevator forces
- Harder to flare



Aerodynamics: Aft CG

- Less stable
- Very light control forces
- Harder stall and spin recovery
- Faster cruise from less drag



PHAK 5-43, PHAK 5-44, PHAK 10-3

Example: C172S Takeoff Setup

- POH sample problem
- Read next higher values

Item	Given
Takeoff weight	2,550 lb
Field pressure altitude	1,500 ft
Temperature	28 C
Wind	12 kt headwind
Field length	3,500 ft
Chart row used	2,000 ft, 30 C

CONDITIONS:

Flaps 10°
Full Throttle Prior to Brake Release
Paved, level, dry runway

Zero Wind
Lift Off: 51 KIAS
Speed at 50 Ft: 56 KIAS

Press Alt In Feet	0°C		10°C		20°C		30°C		40°C	
	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst
S. L.	860	1465	925	1575	995	1690	1070	1810	1150	1945
1000	940	1600	1010	1720	1090	1850	1170	1990	1260	2135
2000	1025	1755	1110	1890	1195	2035	1285	2190	1380	2355
3000	1125	1925	1215	2080	1310	2240	1410	2420	1515	2605
4000	1235	2120	1335	2295	1440	2480	1550	2685	1660	2880
5000	1355	2345	1465	2545	1585	2755	1705	2975	1825	3205
6000	1495	2605	1615	2830	1745	3075	1875	3320	2010	3585
7000	1645	2910	1785	3170	1920	3440	2065	3730	2215	4045
8000	1820	3265	1970	3575	2120	3880	2280	4225	2450	4615

POH 5-3, POH 5-4, POH Fig 5-5

Example: Takeoff with Wind Correction

- 12/9 x 10% = 13% shorter

Distance (ft)	Chart	Minus 13%	Corrected
Ground roll	1,285	167	1,118
Over 50 ft obstacle	2,190	285	1,905

CONDITIONS:

Flaps 10°
Full Throttle Prior to Brake Release
Paved, level, dry runway

Zero Wind
Lift Off: 51 KIAS
Speed at 50 Ft: 56 KIAS

Press Alt In Feet	0°C		10°C		20°C		30°C		40°C	
	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst
S. L.	860	1465	925	1575	995	1690	1070	1810	1150	1945
1000	940	1600	1010	1720	1090	1850	1170	1990	1260	2135
2000	1025	1755	1110	1890	1195	2035	1285	2190	1380	2355
3000	1125	1925	1215	2080	1310	2240	1410	2420	1515	2605
4000	1235	2120	1335	2295	1440	2480	1550	2685	1660	2880
5000	1355	2345	1465	2545	1585	2755	1705	2975	1825	3205
6000	1495	2605	1615	2830	1745	3075	1875	3320	2010	3585
7000	1645	2910	1785	3170	1920	3440	2065	3730	2215	4045
8000	1820	3265	1970	3575	2120	3880	2280	4225	2450	4615

POH 5-4, POH 5-5, POH Fig 5-5

Example: Fuel and Distance to Climb

- 2,000 to 8,000 ft: 2.2 gal
- Hot day: +16% (16 C)
- Corrected fuel: 2.6 gal
- Distance: 18 nm

TIME, FUEL AND DISTANCE TO CLIMB AT 2550 POUNDS

CONDITIONS:

Flaps Up
Full Throttle
Standard Temperature

PRESS ALT FT	CLIMB SPEED KIAS	RATE OF CLIMB FPM	FROM SEA LEVEL		
			TIME IN MIN	FUEL USED GAL	DIST NM
S.L.	74	730	0	0.0	0
1000	73	695	1	0.4	2
2000	73	655	3	0.8	4
3000	73	620	4	1.2	6
4000	73	600	6	1.5	8
5000	73	550	8	1.9	10
6000	73	505	10	2.2	13
7000	73	455	12	2.6	16
8000	72	410	14	3.0	19
9000	72	360	17	3.4	22
10,000	72	315	20	3.9	27
11,000	72	265	24	4.4	32
12,000	72	220	28	5.0	38

NOTE

- Add 1.4 gallons of fuel for engine start, taxi and takeoff allowance.
- Mixture leaned above 3000 feet elevation for maximum RPM.
- Increase time, fuel and distance by 10% for each 10°C above standard temperature.
- Distances shown are based on zero wind.

POH 5-6, POH Fig 5-7

Example: Cruise Performance

- 8,000 ft, 20 C above standard
- 2,600 RPM: 64% power
- 117 KTAS at 8.9 GPH
- 10 kt headwind: 107 kt GS

CRUISE PERFORMANCE

CONDITIONS:
2550 Pounds
Recommended Lean Mixture At All Altitudes (Refer to Section 4, Cruise)

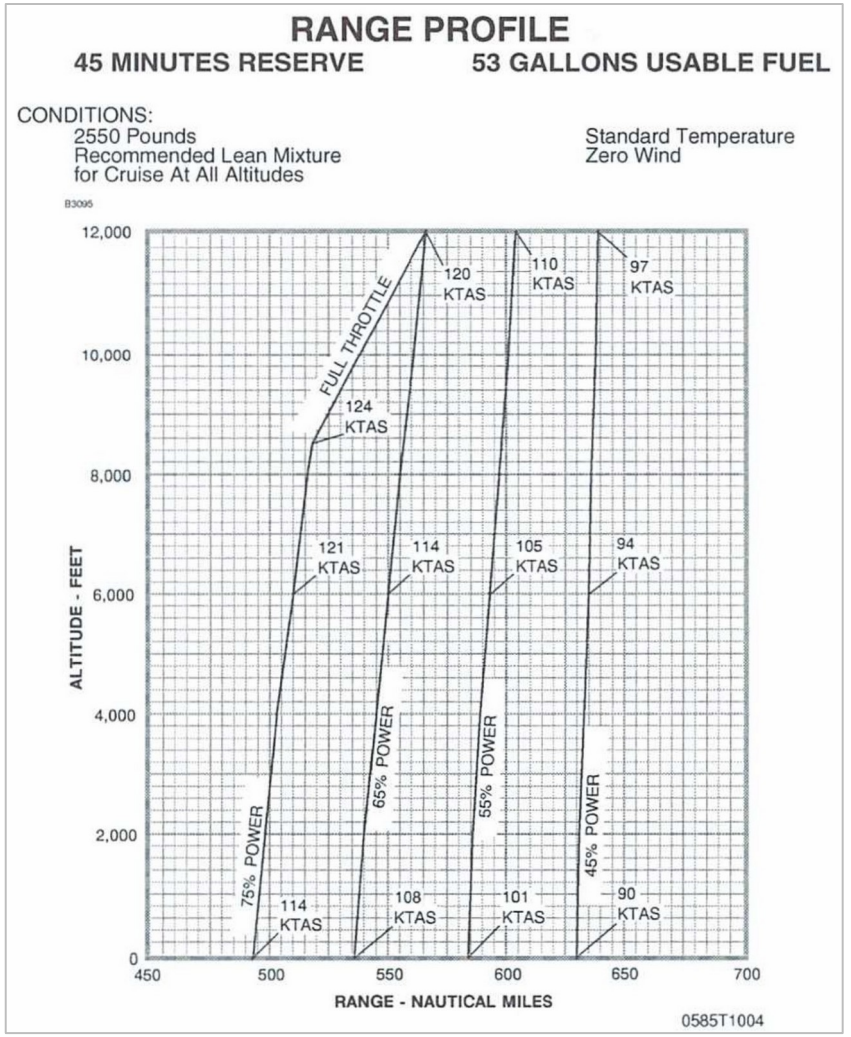
PRESS ALT FT	RPM	20°C BELOW STANDARD TEMP			STANDARD TEMPERATURE			20°C ABOVE STANDARD TEMP		
		% MCP	KTAS	GPH	% MCP	KTAS	GPH	% MCP	KTAS	GPH
2000	2550	83	117	11.1	77	118	10.5	72	117	9.9
	2500	78	115	10.6	73	115	9.9	68	115	9.4
	2400	69	111	9.6	64	110	9.0	60	109	8.5
	2300	61	105	8.6	57	104	8.1	53	102	7.7
	2200	53	99	7.7	50	97	7.3	47	95	6.9
	2100	47	92	6.9	44	90	6.6	42	89	6.3
4000	2600	83	120	11.1	77	120	10.4	72	119	9.8
	2550	79	118	10.6	73	117	9.9	68	117	9.4
	2500	74	115	10.1	69	115	9.5	64	114	8.9
	2400	65	110	9.1	61	109	8.5	57	107	8.1
	2300	58	104	8.2	54	102	7.7	51	101	7.3
	2200	51	98	7.4	48	96	7.0	45	94	6.7
	2100	45	91	6.6	42	89	6.4	40	87	6.1
6000	2650	83	122	11.1	77	122	10.4	72	121	9.8
	2600	78	120	10.6	73	119	9.9	68	118	9.4
	2500	70	115	9.6	65	114	9.0	60	112	8.5
	2400	62	109	8.6	57	108	8.2	54	106	7.7
	2300	54	103	7.8	51	101	7.4	48	99	7.0
	2200	48	96	7.1	45	94	6.7	43	92	6.4

POH 5-5, POH 5-7, POH Fig 5-8

Example: Total Fuel Required

- 360 nm trip, 53 gal usable

Leg	Gallons
Start, taxi, takeoff	1.4
Climb (18 nm)	2.6
Cruise (342 nm, 3.2 hr)	28.5
Reserve (45 min)	6.7
Total required	**39.2**



POH 5-6, POH 5-7, POH Fig 5-9

Example: Landing Distance

- Destination: 2,000 ft, 25 C
- Read at 2,000 ft, 30 C
- Ground roll: 650 ft
- Over 50 ft obstacle: 1,455 ft

CONDITIONS:

Flaps Full
Power Off
Maximum Braking

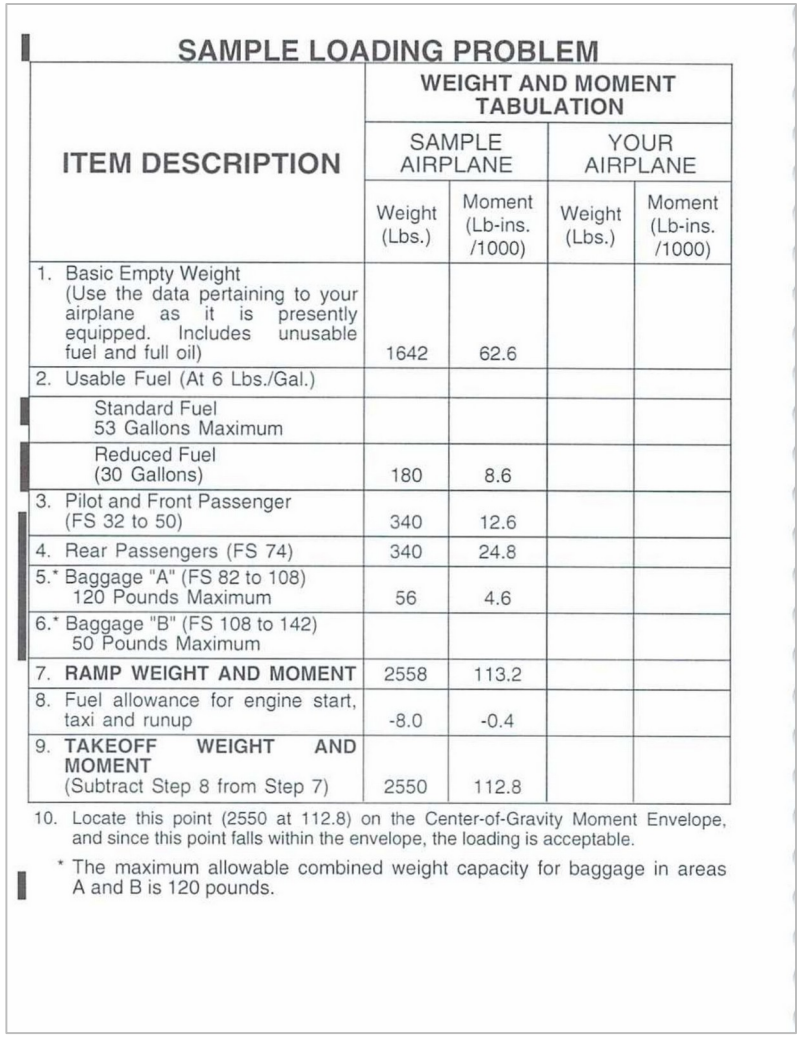
Paved, level, dry runway
Zero Wind
Speed at 50 Ft: 61 KIAS

Press Alt In Feet	0°C		10°C		20°C		30°C		40°C	
	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst	Grnd Roll Ft	Total Ft To Clear 50 Ft Obst
S. L.	545	1290	565	1320	585	1350	605	1380	625	1415
1000	565	1320	585	1350	605	1385	625	1420	650	1450
2000	585	1355	610	1385	630	1420	650	1455	670	1490
3000	610	1385	630	1425	655	1460	675	1495	695	1530
4000	630	1425	655	1460	675	1495	700	1535	725	1570
5000	655	1460	680	1500	705	1535	725	1575	750	1615
6000	680	1500	705	1540	730	1580	755	1620	780	1660
7000	705	1545	730	1585	760	1625	785	1665	810	1705
8000	735	1585	760	1630	790	1670	815	1715	840	1755

POH 5-8, POH Fig 5-11

Example: C172S Sample Loading

Item	Weight (lb)	Moment /1000
Basic empty weight	1,642	62.6
Fuel (30 gal)	180	8.6
Pilot and front pass.	340	12.6
Rear passengers	340	24.8
Baggage A	56	4.6
Ramp	2,558	113.2
Start and taxi fuel	-8	-0.4
Takeoff	**2,550**	**112.8**



SAMPLE LOADING PROBLEM

ITEM DESCRIPTION	WEIGHT AND MOMENT TABULATION			
	SAMPLE AIRPLANE		YOUR AIRPLANE	
	Weight (Lbs.)	Moment (Lb-ins. /1000)	Weight (Lbs.)	Moment (Lb-ins. /1000)
1. Basic Empty Weight (Use the data pertaining to your airplane as it is presently equipped. Includes unusable fuel and full oil)	1642	62.6		
2. Usable Fuel (At 6 Lbs./Gal.)				
Standard Fuel 53 Gallons Maximum				
Reduced Fuel (30 Gallons)	180	8.6		
3. Pilot and Front Passenger (FS 32 to 50)	340	12.6		
4. Rear Passengers (FS 74)	340	24.8		
5.* Baggage "A" (FS 82 to 108) 120 Pounds Maximum	56	4.6		
6.* Baggage "B" (FS 108 to 142) 50 Pounds Maximum				
7. **RAMP WEIGHT AND MOMENT**	2558	113.2		
8. Fuel allowance for engine start, taxi and runup	-8.0	-0.4		
9. **TAKEOFF WEIGHT AND MOMENT** (Subtract Step 8 from Step 7)	2550	112.8		

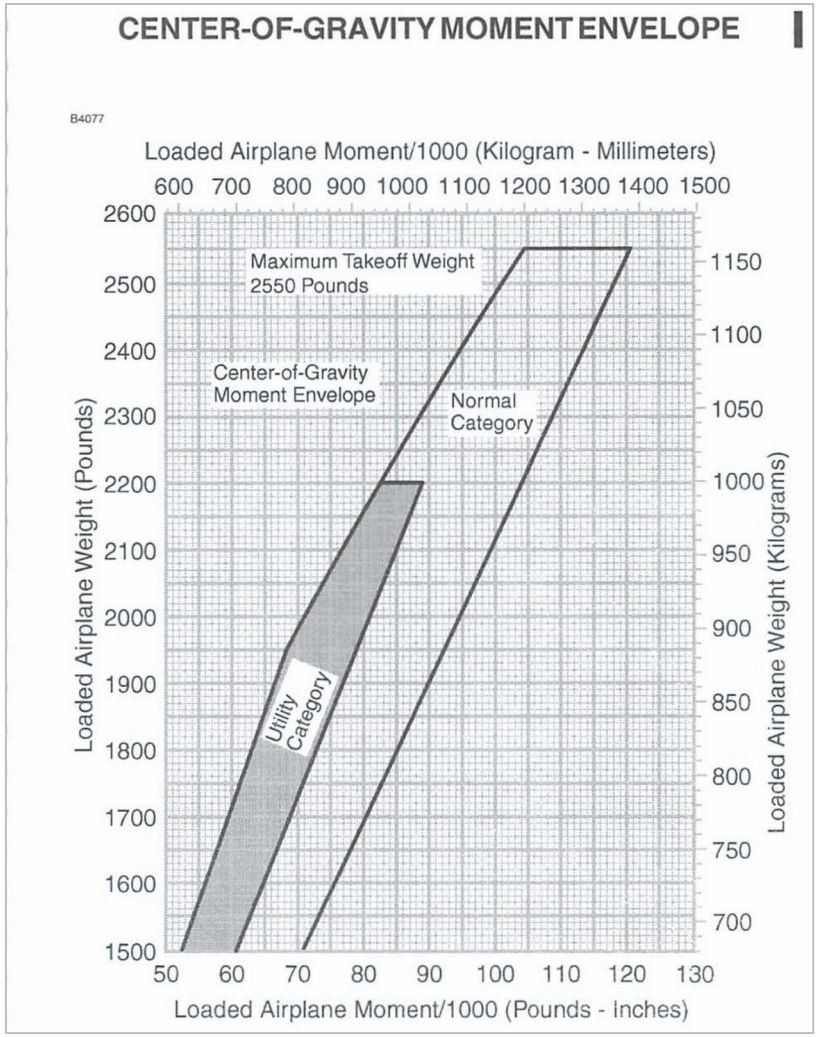
10. Locate this point (2550 at 112.8) on the Center-of-Gravity Moment Envelope, and since this point falls within the envelope, the loading is acceptable.

* The maximum allowable combined weight capacity for baggage in areas A and B is 120 pounds.

POH 6-10, POH Fig 6-3

Example: Plot on the Envelope

- Plot 2,550 lb at 112.8
- Inside envelope: approved
- CG: 112,800 / 2,550 = 44.2 in
- Limits: 41.0 to 47.3 in

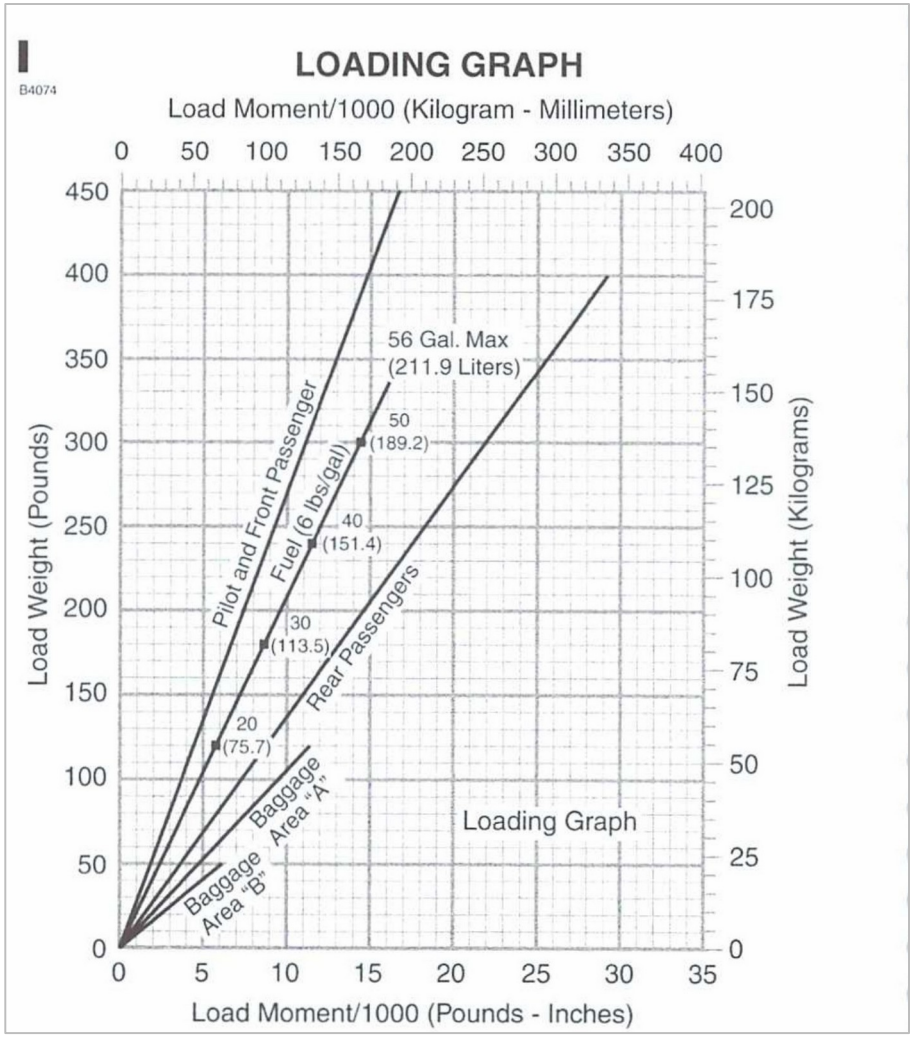


POH 6-10, POH Fig 6-7, POH 2-8

Check Every Phase: Fuel Burn

- Fuel arm: 48.0 in
- Burn 15 gal (90 lb)

Phase	Weight	Moment /1000	CG (in)
Takeoff	2,550	112.8	44.2
Fuel burned	-90	-4.3	
Landing	**2,460**	**108.5**	**44.1**



TCDS 3A12 p30, PHAK 10-11, POH 2-8, POH Fig 6-4

Fixing an Out-of-Limits CG

- Weight moved / total weight
- = CG change / distance moved
- PHAK: move 65 lb forward
- Recheck after every change

Example

Given:

Aircraft total weight . . . . . 7,800 lb
CG station. . . . . 81.5 in
Aft CG limit . . . . . 80.5 in

Determine how much cargo must be shifted from the aft cargo compartment at station 150 to the forward cargo compartment at station 30 to move the CG to exactly the aft limit.

Solution:

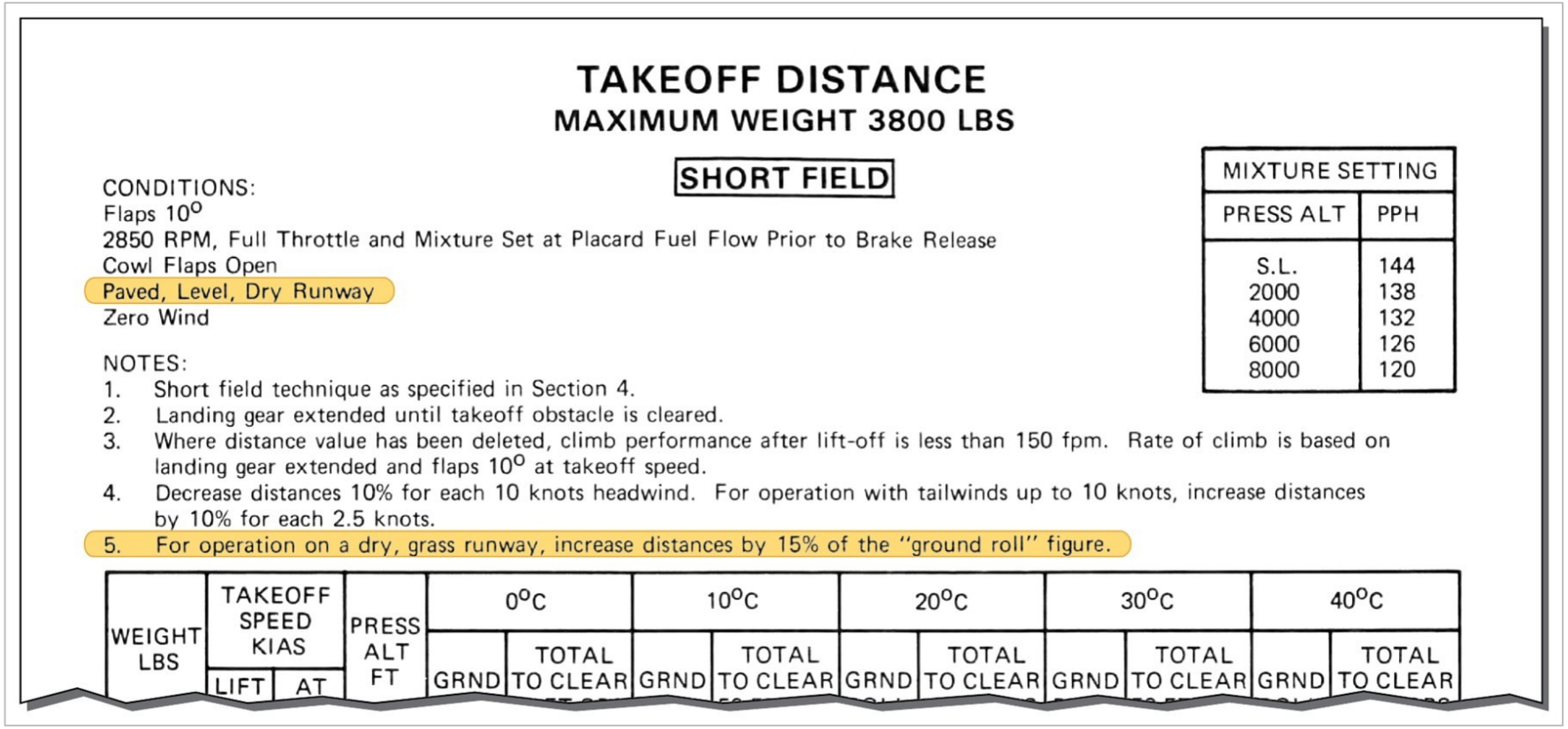
$$\frac{\text{Weight to be shifted}}{\text{Total weight}} = \frac{\Delta CG}{\text{Distance weight is shifted}}$$

$$\frac{\text{Weight to be shifted}}{7{,}800\ \text{lb}} = \frac{1.0\ \text{in}}{120\ \text{in}}$$

$$\text{Weight to be shifted} = 65\ \text{lb}$$

Risk: Charts vs. Real Life

- Expect 10% or more variation
- Use conservative chart values
- Set an abort point
- Scenario: hot day, short strip



TAKEOFF DISTANCE
MAXIMUM WEIGHT 3800 LBS

SHORT FIELD

CONDITIONS:
Flaps 10°
2850 RPM, Full Throttle and Mixture Set at Placard Fuel Flow Prior to Brake Release
Cowl Flaps Open
Paved, Level, Dry Runway
Zero Wind

MIXTURE SETTING	
PRESS ALT	PPH
S.L.	144
2000	138
4000	132
6000	126
8000	120

NOTES:
1. Short field technique as specified in Section 4.
2. Landing gear extended until takeoff obstacle is cleared.
3. Where distance value has been deleted, climb performance after lift-off is less than 150 fpm. Rate of climb is based on landing gear extended and flaps 10° at takeoff speed.
4. Decrease distances 10% for each 10 knots headwind. For operation with tailwinds up to 10 knots, increase distances by 10% for each 2.5 knots.
5. For operation on a dry, grass runway, increase distances by 15% of the "ground roll" figure.

WEIGHT LBS	TAKEOFF SPEED KIAS		PRESS ALT FT	0°C		10°C		20°C		30°C		40°C	
	LIFT	AT		GRND	TOTAL TO CLEAR	GRND	TOTAL TO CLEAR	GRND	TOTAL TO CLEAR	GRND	TOTAL TO CLEAR	GRND	TOTAL TO CLEAR

POH 5-3, POH 5-4, PHAK 11-19

Risk: Limits and Overweight

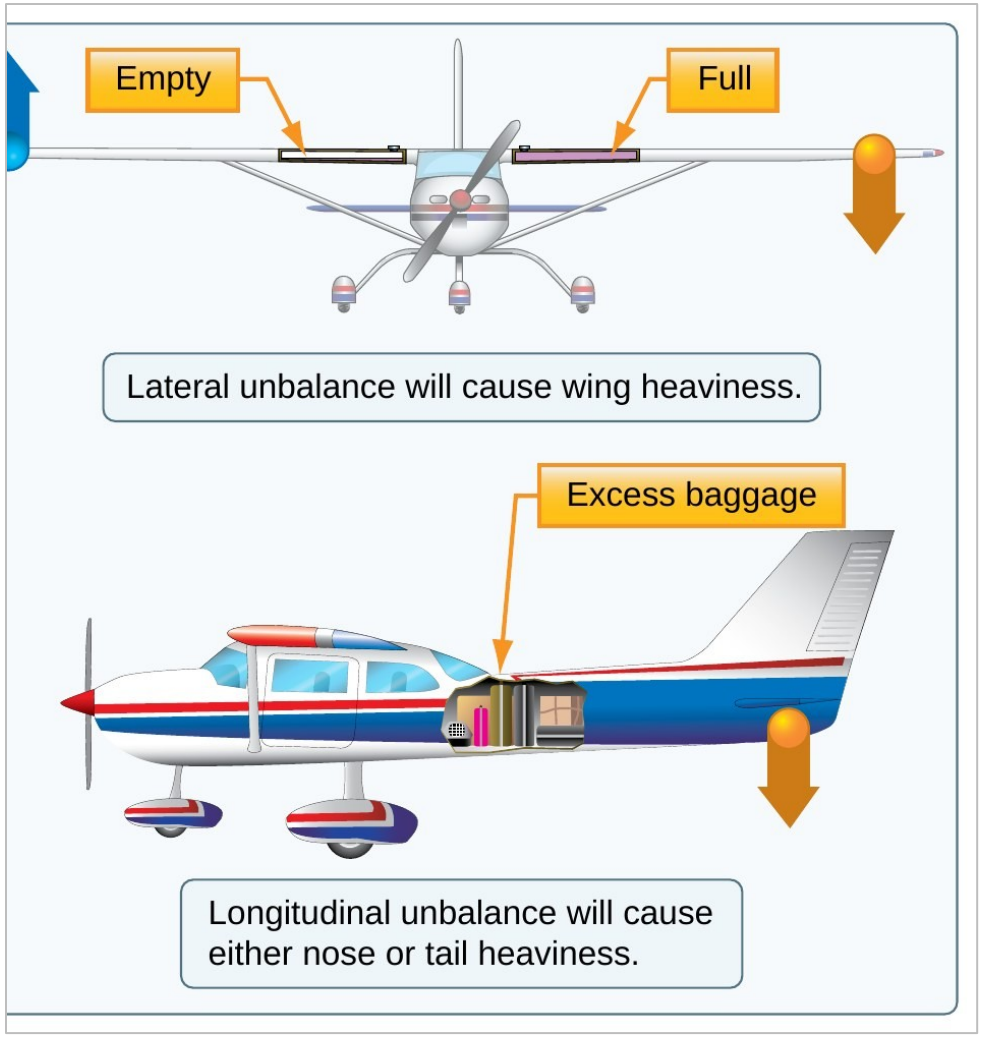
- POH limits are mandatory
- Overweight: poor climb, high stall
- Structure not proven above max
- Scenario: four adults, full tanks

SYMBOL	SPEED	KCAS	KIAS	REMARKS
V_{NE}	Never Exceed Speed	160	163	Do not exceed this speed in any operation.
V_{NO}	Maximum Structural Cruising Speed	126	129	Do not exceed this speed except in smooth air, and then only with caution.
V_A	Maneuvering Speed: 2550 Pounds 2200 Pounds 1900 Pounds	 102 95 88	 105 98 90	Do not make full or abrupt control movements above this speed.
V_{FE}	Maximum Flap Extended Speed: Flaps 10° Flaps 10° to FULL	 107 85	 110 85	Do not exceed this speed with flaps down.
-----	Maximum Window Open Speed	160	163	Do not exceed this speed with windows open.

PHAK 10-1, PHAK 10-2, PHAK 10-4, POH 2-4

Risk: CG Limits and Shifting Weight

- Out of CG: control trouble
- Aft CG: stall and spin danger
- Moved bags move the CG
- Scenario: heavy bags in back



Common Errors

- Using the sample empty weight
- Skipping the chart notes
- No temperature correction
- Forgetting taxi fuel

TIME, FUEL AND DISTANCE TO CLIMB AT 2550 POUNDS

CONDITIONS:

Flaps Up
Full Throttle
Standard Temperature

PRESS ALT FT	CLIMB SPEED KIAS	RATE OF CLIMB FPM	FROM SEA LEVEL		
			TIME IN MIN	FUEL USED GAL	DIST NM
S.L.	74	730	0	0.0	0
1000	73	695	1	0.4	2
2000	73	655	3	0.8	4
3000	73	620	4	1.2	6
4000	73	600	6	1.5	8
5000	73	550	8	1.9	10
6000	73	505	10	2.2	13
7000	73	455	12	2.6	16
8000	72	410	14	3.0	19
9000	72	360	17	3.4	22
10,000	72	315	20	3.9	27
11,000	72	265	24	4.4	32
12,000	72	220	28	5.0	38

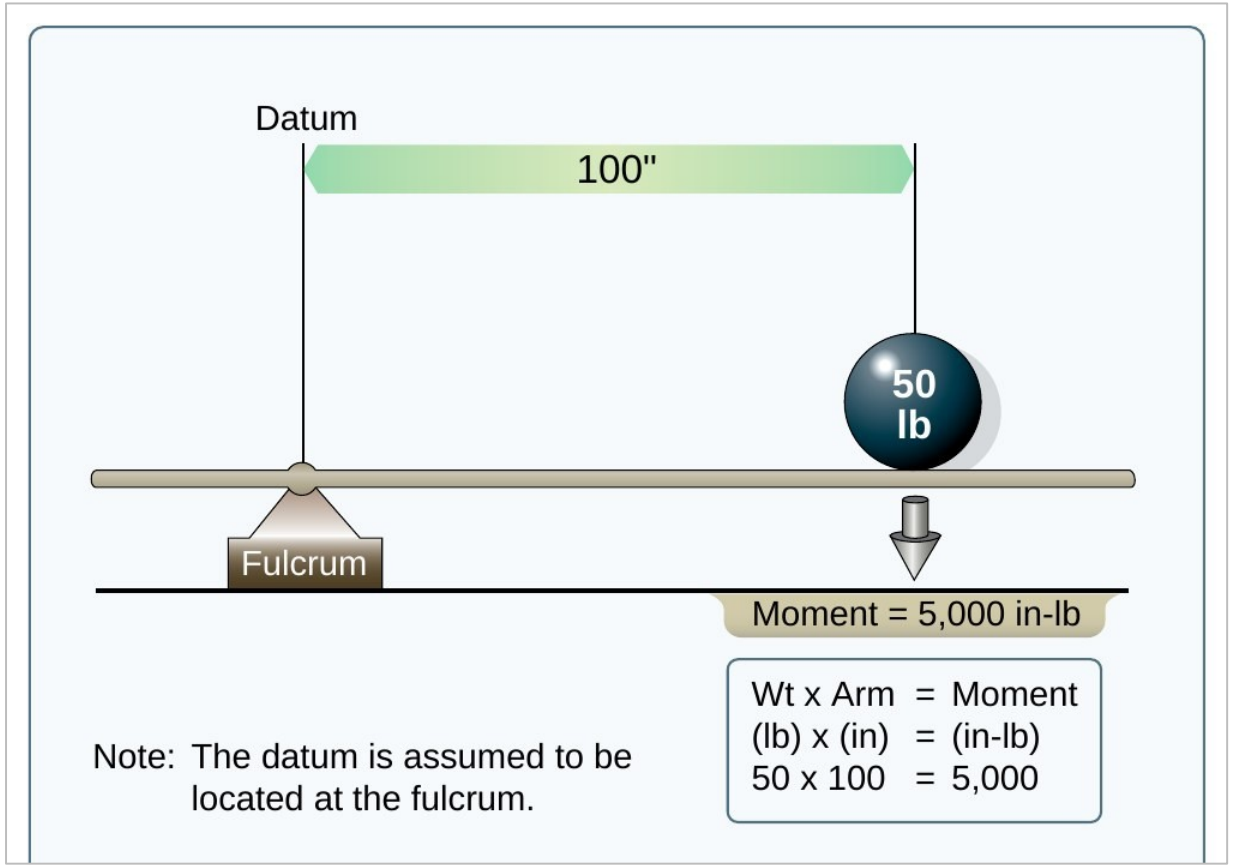
NOTE

- Add 1.4 gallons of fuel for engine start, taxi and takeoff allowance.
- Mixture leaned above 3000 feet elevation for maximum RPM.
- Increase time, fuel and distance by 10% for each 10°C above standard temperature.
- Distances shown are based on zero wind.

PHAK 10-6, POH 5-4, POH 5-6, POH 6-10, POH Fig 5-5, POH Fig 5-7

Review Questions

- What raises density altitude?
- Where is the C172S datum?
- How do you find CG?
- 18 kt headwind: what change?



PHAK 11-2, PHAK 10-7, TCDS 3A12 p30, POH Fig 5-5

References

- FAA-S-ACS-25, Task II.F
- PHAK Ch 5, 10, 11
- C172S POH Sec 2, 5, 6
- TCDS 3A12 Rev 89

PERFORMANCE

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See lesson plan for full reference list